SVM

Linear

	x'_1	x'_2
x'_j	2	1
	4	1
	3	1

y_i		x_i					
1	x_1	1	2	1			
-1	x_2	0	4	3			
1	x_3	0	3	0			
-1	x_4	0	3	1			
1	x_5	5	0	3			
-1	x_6	1	4	3			

Kernel Matrix

$K(x_i, x'_j)$

a_i	2	0	1	1	1	1	b	
$a_i\, y_i$							2	
sign								

Decision Boundary

$b + \sum a_i\, y_i\, K$

RBF

x'_j
2
4
3

$-\gamma\|\|^2$	$e^{\square}$
-0.5	0.6
-0.4	0.67
-0.3	0.7
-0.2	0.8
-0.1	0.9

1
1
1

L2 distance $||^2$

x_i			$(x - x')^2$			$\sum$	$\sqrt{\ }$	$-\gamma\|\|^2$ (-0.1)	$e^{\square}$									
1	2	1																
0	4	3																
0	3	0																
0	3	1																
5	0	3																
1	4	3																

a_i	20	10	10	10	0	10	b		
$a_i\, y_i$							-1		
sign									

SVM

Linear

	x'_1	x'_2
x'_j	2	1
	4	1
	3	1

y_i		x_i					
1	x_1	1	2	1	13	
-1	x_2	0	4	3	25	
1	x_3	0	3	0	12	
-1	x_4	0	3	1	15	
1	x_5	5	0	3	19	
-1	x_6	1	4	3	27	

Kernel Matrix

$K(x_i, x'_j)$

a_i	2	0	1	1	1	1	b
$a_i\, y_i$							2
sign							

Decision Boundary

$b + \sum a_i\, y_i\, K$

RBF

x'_j	2
	4
	3

L2 distance $||^2$

$-\gamma||^2$	$e^{\square}$
-0.5	0.6
-0.4	0.67
-0.3	0.7
-0.2	0.8
-0.1	0.9

1
1
1

x_i			$(x - x')^2$			$\sum$	$\sqrt{}$	$-\gamma||^2$ (-0.1)	$e^{\square}$						
1	2	1													
0	4	3													
0	3	0													
0	3	1													
5	0	3													
1	4	3													

a_i	20	10	10	10	0	10	b
$a_i\, y_i$							-1
sign							

SVM

Linear

	x'_1	x'_2
x'_j	2	1
	4	1
	3	1

y_i		x_i					
1	x_1	1	2	1	13	2
-1	x_2	0	4	3	25	7
1	x_3	0	3	0	12	3
-1	x_4	0	3	1	15	4
1	x_5	5	0	3	19	8
-1	x_6	1	4	3	27	8

Kernel Matrix

$K(x_i, x'_j)$

a_i	2	0	1	1	1	1	b
$a_i\ y_i$							2
sign							

Decision Boundary

$b + \sum a_i\ y_i\ K$

RBF

x'_j	2
	4
	3

L2 distance $||^2$

$-\gamma||^2$	$e^{\square}$
-0.5	0.6
-0.4	0.67
-0.3	0.7
-0.2	0.8
-0.1	0.9

1
1
1

x_i			$(x - x')^2$			$\sum$	$\sqrt{\ }$	$-\gamma||^2$ (-0.1)	$e^{\square}$						
1	2	1													
0	4	3													
0	3	0													
0	3	1													
5	0	3													
1	4	3													

a_i	20	10	10	10	0	10	b
$a_i\ y_i$							-1
sign							

SVM

Linear

	x'_1	x'_2
x'_j	2	1
	4	1
	3	1

y_i		x_i					
1	x_1	1	2	1	13	2
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1	x_3	0	3	0	12	3
-1	x_4	0	3	1	15	4
1	x_5	5	0	3	19	8
-1	x_6	1	4	3	27	8

Kernel Matrix

$K(x_i, x'_j)$

							b
a_i	2	0	1	1	1	1	
$a_i\, y_i$	2	0	1	-1	1	-1	2
sign							

Decision Boundary

$b + \sum a_i\, y_i\, K$

RBF

x'_j

2
4
3

L2 distance $||^2$

$-\gamma||^2$	$e^{\square}$
-0.5	0.6
-0.4	0.67
-0.3	0.7
-0.2	0.8
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1
1
1

x_i			$(x - x')^2$			$\sum$	$\sqrt{\ }$	$-\gamma||^2$ (-0.1)	$e^{\square}$						
1	2	1													
0	4	3													
0	3	0													
0	3	1													
5	0	3													
1	4	3													

							b
a_i	20	10	10	10	0	10	
$a_i\, y_i$							-1
sign							

SVM

Linear

	x'_1	x'_2
x'_j	2	1
	4	1
	3	1

y_i		x_i					
1	x_1	1	2	1	13	2
-1	x_2	0	4	3	25	7
1	x_3	0	3	0	12	3
-1	x_4	0	3	1	15	4
1	x_5	5	0	3	19	8
-1	x_6	1	4	3	27	8

Kernel Matrix

$K(x_i, x'_j)$

							b		
a_i	2	0	1	1	1	1			
$a_i\, y_i$	2	0	1	-1	1	-1	2	17	5
sign									

Decision Boundary

$b + \sum a_i\, y_i\, K$

RBF

x'_j	2
	4
	3

L2 distance $||^2$

$-\gamma||^2$	$e^{\square}$
-0.5	0.6
-0.4	0.67
-0.3	0.7
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-0.1	0.9

1
1
1

x_i			$(x - x')^2$			$\sum$	$\sqrt{\ }$	$-\gamma||^2$ (-0.1)	$e^{\square}$								
1	2	1															
0	4	3															
0	3	0															
0	3	1															
5	0	3															
1	4	3															

							b		
a_i	20	10	10	10	0	10			
$a_i\, y_i$							-1		
sign									

SVM

Linear

	x'_1	x'_2
x'_j	2	1
	4	1
	3	1

y_i		x_i					
1	x_1	1	2	1	13	2
-1	x_2	0	4	3	25	7
1	x_3	0	3	0	12	3
-1	x_4	0	3	1	15	4
1	x_5	5	0	3	19	8
-1	x_6	1	4	3	27	8

Kernel Matrix

$K(x_i, x'_j)$

							b		
a_i	2	0	1	1	1	1			
$a_i\, y_i$	2	0	1	-1	1	-1	2	17	5
sign								+	+

Decision Boundary

$b + \sum a_i\, y_i\, K$

RBF

x'_j	2
	4
	3

L2 distance $||^2$

$-\gamma||^2$	$e^{\square}$
-0.5	0.6
-0.4	0.67
-0.3	0.7
-0.2	0.8
-0.1	0.9

1
1
1

x_i			$(x - x')^2$			$\sum$	$\sqrt{\ }$	$-\gamma||^2$ (-0.1)	$e^{\square}$							
1	2	1														
0	4	3														
0	3	0														
0	3	1														
5	0	3														
1	4	3														

							b		
a_i	20	10	10	10	0	10			
$a_i\, y_i$							-1		
sign									

SVM

Linear

	x'_1	x'_2
x'_j	2	1
	4	1
	3	1

y_i		x_i					
1	x_1	1	2	1	13	2
-1	x_2	0	4	3	25	7
1	x_3	0	3	0	12	3
-1	x_4	0	3	1	15	4
1	x_5	5	0	3	19	8
-1	x_6	1	4	3	27	8

Kernel Matrix

$K(x_i, x'_j)$

							b		
a_i	2	0	1	1	1	1			
$a_i\, y_i$	2	0	1	-1	1	-1	2	17	5
sign								+	+

Decision Boundary

$b + \sum a_i\, y_i\, K$

RBF

	x'_1	x'_2
x'_j	2	1
	4	1
	3	1

L2 distance $||^2$

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-0.5	0.6
-0.4	0.67
-0.3	0.7
-0.2	0.8
-0.1	0.9

x_i			$(x - x')^2$			$\sum$	$\sqrt{\ }$	$-\gamma||^2$ (-0.1)	$e^{\square}$							
1	2	1	1	4	4											
0	4	3	4	0	0											
0	3	0	0	1	0											
0	3	1	4	1	4											
5	0	3	9	16	0											
1	4	3	1	0	0											

							b		
a_i	20	10	10	10	0	10			
$a_i\, y_i$							-1		
sign									

SVM

Linear

	x'_1	x'_2
x'_j	2	1
	4	1
	3	1

y_i		x_i					
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1	x_5	5	0	3	19	8
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Kernel Matrix

$K(x_i, x'_j)$

							b		
a_i	2	0	1	1	1	1			
$a_i\, y_i$	2	0	1	-1	1	-1	2	→ 17	5
sign								+	+

Decision Boundary

$b + \sum a_i\, y_i\, K$

RBF

	x'_1	x'_2
x'_j	2	1
	4	1
	3	1

L2 distance $||^2$

$-\gamma||^2$	$e^{\square}$
-0.5	0.6
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-0.2	0.8
-0.1	0.9

x_i			$(x - x')^2$			$\sum$	$\sqrt{\ }$	$-\gamma||^2$ (-0.1)	$e^{\square}$									
1	2	1	1	4	4	9												
0	4	3	4	0	0	4												
0	3	0	0	1	0	1												
0	3	1	4	1	4	9												
5	0	3	9	16	0	25												
1	4	3	1	0	0	1												

							b		
a_i	20	10	10	10	0	10			
$a_i\, y_i$							-1		
sign									

SVM

Linear

	x'_1	x'_2
x'_j	2	1
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y_i		x_i					
1	x_1	1	2	1	13	2
-1	x_2	0	4	3	25	7
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Kernel Matrix

$K(x_i, x'_j)$

							b		
a_i	2	0	1	1	1	1			
$a_i\, y_i$	2	0	1	-1	1	-1	2	17	5
sign								+	+

Decision Boundary

$b + \sum a_i\, y_i\, K$

RBF

	x'_1	x'_2
x'_j	2	1
	4	1
	3	1

L2 distance $||^2$

$-\gamma||^2$	$e^{\square}$
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-0.4	0.67
-0.3	0.7
-0.2	0.8
-0.1	0.9

x_i			$(x - x')^2$			$\sum$	$\sqrt{\ }$	$-\gamma||^2$ (-0.1)	$e^{\square}$						
1	2	1	1	4	4	9	3								
0	4	3	4	0	0	4	2								
0	3	0	0	1	0	1	1								
0	3	1	4	1	4	9	3								
5	0	3	9	16	0	25	5								
1	4	3	1	0	0	1	1								

							b		
a_i	20	10	10	10	0	10			
$a_i\, y_i$							-1		
sign									

SVM

Linear

	x'_1	x'_2
x'_j	2	1
	4	1
	3	1

y_i		x_i					
1	x_1	1	2	1	13	2
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1	x_3	0	3	0	12	3
-1	x_4	0	3	1	15	4
1	x_5	5	0	3	19	8
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Kernel Matrix

$K(x_i, x'_j)$

							b		
a_i	2	0	1	1	1	1			
$a_i\, y_i$	2	0	1	-1	1	-1	2	17	5
sign								+	+

Decision Boundary

$b + \sum a_i\, y_i\, K$

RBF

	x'_1	x'_2
x'_j	2	1
	4	1
	3	1

L2 distance $||^2$

$-\gamma||^2$	$e^{\square}$
-0.5	0.6
-0.4	0.67
-0.3	0.7
-0.2	0.8
-0.1	0.9

x_i			$(x - x')^2$			$\sum$	$\sqrt{\ }$	$-\gamma||^2$ (-0.1)	$e^{\square}$						
1	2	1	1	4	4	9	3	-0.3							
0	4	3	4	0	0	4	2	-0.2							
0	3	0	0	1	0	1	1	-0.1							
0	3	1	4	1	4	9	3	-0.3							
5	0	3	9	16	0	25	5	-0.5							
1	4	3	1	0	0	1	1	-0.1							

							b		
a_i	20	10	10	10	0	10			
$a_i\, y_i$							-1		
sign									

SVM

Linear

	x'_1	x'_2
x'_j	2	1
	4	1
	3	1

y_i		x_i					
1	x_1	1	2	1	13	2
-1	x_2	0	4	3	25	7
1	x_3	0	3	0	12	3
-1	x_4	0	3	1	15	4
1	x_5	5	0	3	19	8
-1	x_6	1	4	3	27	8

Kernel Matrix

$K(x_i, x'_j)$

							b		
a_i	2	0	1	1	1	1			
$a_i\, y_i$	2	0	1	-1	1	-1	2	17	5
sign								+	+

Decision Boundary

$b + \sum a_i\, y_i\, K$

RBF

	x'_1	x'_2
x'_j	2	1
	4	1
	3	1

L2 distance $||^2$

$-\gamma||^2$	$e^{\square}$
-0.5	0.6
-0.4	0.67
-0.3	0.7
-0.2	0.8
-0.1	0.9

x_i			$(x - x')^2$			$\sum$	$\sqrt{\ }$	$-\gamma||^2$ (-0.1)	$e^{\square}$
1	2	1	1	4	4	9	3	-0.3	0.7
0	4	3	4	0	0	4	2	-0.2	0.8
0	3	0	0	1	0	1	1	-0.1	0.9
0	3	1	4	1	4	9	3	-0.3	0.7
5	0	3	9	16	0	25	5	-0.5	0.6
1	4	3	1	0	0	1	1	-0.1	0.9

							b		
a_i	20	10	10	10	0	10			
$a_i\, y_i$							-1		
sign									

SVM

Linear

	x'_1	x'_2
x'_j	2	1
	4	1
	3	1

y_i		x_i					
1	x_1	1	2	1	13	2
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1	x_5	5	0	3	19	8
-1	x_6	1	4	3	27	8

Kernel Matrix

$K(x_i, x'_j)$

							b		
a_i	2	0	1	1	1	1			
$a_i\, y_i$	2	0	1	-1	1	-1	2	17	5
sign								+	+

Decision Boundary

$b + \sum a_i\, y_i\, K$

RBF

	x'_1	x'_2
x'_j	2	1
	4	1
	3	1

L2 distance $||^2$

$-\gamma||^2$	$e^{\square}$
-0.5	0.6
-0.4	0.67
-0.3	0.7
-0.2	0.8
-0.1	0.9

x_i			$(x - x')^2$			$\sum$	$\sqrt{\ }$	$-\gamma||^2$ (-0.1)	$e^{\square}$						
1	2	1	1	4	4	9	3	-0.3	0.7	0	1	0			
0	4	3	4	0	0	4	2	-0.2	0.8	1	9	4			
0	3	0	0	1	0	1	1	-0.1	0.9	1	4	1			
0	3	1	4	1	4	9	3	-0.3	0.7	1	4	0			
5	0	3	9	16	0	25	5	-0.5	0.6	16	1	4			
1	4	3	1	0	0	1	1	-0.1	0.9	0	9	4			

							b		
a_i	20	10	10	10	0	10			
$a_i\, y_i$							-1		
sign									

SVM

Linear

	x'_1	x'_2
x'_j	2	1
	4	1
	3	1

y_i		x_i					
1	x_1	1	2	1	13	2
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-1	x_4	0	3	1	15	4
1	x_5	5	0	3	19	8
-1	x_6	1	4	3	27	8

Kernel Matrix

$K(x_i, x'_j)$

							b		
a_i	2	0	1	1	1	1			
$a_i\, y_i$	2	0	1	-1	1	-1	2	17	5
sign								+	+

Decision Boundary

$b + \sum a_i\, y_i\, K$

RBF

	x'_1	x'_2
x'_j	2	1
	4	1
	3	1

L2 distance $||^2$

$-\gamma||^2$	$e^{\square}$
-0.5	0.6
-0.4	0.67
-0.3	0.7
-0.2	0.8
-0.1	0.9

x_i			$(x - x')^2$			$\sum$	$\sqrt{\ }$	$-\gamma||^2$ (-0.1)	$e^{\square}$						
1	2	1	1	4	4	9	3	-0.3	0.7	0	1	0	1		
0	4	3	4	0	0	4	2	-0.2	0.8	1	9	4	14		
0	3	0	0	1	0	1	1	-0.1	0.9	1	4	1	6		
0	3	1	4	1	4	9	3	-0.3	0.7	1	4	0	5		
5	0	3	9	16	0	25	5	-0.5	0.6	16	1	4	21		
1	4	3	1	0	0	1	1	-0.1	0.9	0	9	4	13		

							b		
a_i	20	10	10	10	0	10			
$a_i\, y_i$							-1		
sign									

SVM

Linear

	x'_1	x'_2
x'_j	2	1
	4	1
	3	1

y_i		x_i					
1	x_1	1	2	1	13	2
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Kernel Matrix

$K(x_i, x'_j)$

							b		
a_i	2	0	1	1	1	1			
$a_i\, y_i$	2	0	1	-1	1	-1	2	17	5
sign								+	+

Decision Boundary

$b + \sum a_i\, y_i\, K$

RBF

x'_j	2	4	3															
x_i	1	2	1	$(x - x')^2$			$\sum$	$\sqrt{\ }$	$-\gamma\|\|^2$ (-0.1)	$e^{\square}$								
	1	4	4	1	4	4	9	3	-0.3	0.7	0	1	0	1	1		
	0	4	3	4	0	0	4	2	-0.2	0.8	1	9	4	14	3		
	0	3	0	0	1	0	1	1	-0.1	0.9	1	4	1	6	2		
	0	3	1	4	1	4	9	3	-0.3	0.7	1	4	0	5	2		
	5	0	3	9	16	0	25	5	-0.5	0.6	16	1	4	21	4		
	1	4	3	1	0	0	1	1	-0.1	0.9	0	9	4	13	3		

$-\gamma\|\|^2$	$e^{\square}$
-0.5	0.6
-0.4	0.67
-0.3	0.7
-0.2	0.8
-0.1	0.9

L2 distance $||^2$

a_i	20	10	10	10	0	10	b		
$a_i\, y_i$							-1		
sign									

SVM

Linear

	x'_1	x'_2
x'_j	2	1
	4	1
	3	1

y_i		x_i					
1	x_1	1	2	1	13	2
-1	x_2	0	4	3	25	7
1	x_3	0	3	0	12	3
-1	x_4	0	3	1	15	4
1	x_5	5	0	3	19	8
-1	x_6	1	4	3	27	8

Kernel Matrix

$K(x_i, x'_j)$

							b		
a_i	2	0	1	1	1	1			
$a_i\, y_i$	2	0	1	-1	1	-1	2	17	5
sign								+	+

Decision Boundary

$b + \sum a_i\, y_i\, K$

RBF

	x'_1	x'_2
x'_j	2	1
	4	1
	3	1

L2 distance $||^2$

$-\gamma||^2$	$e^{\square}$
-0.5	0.6
-0.4	0.67
-0.3	0.7
-0.2	0.8
-0.1	0.9

x_i			$(x - x')^2$			$\sum$	$\sqrt{\ }$	$-\gamma||^2$ (-0.1)	$e^{\square}$				$\sqrt{\ }$	$-\gamma||^2$	$e^{\square}$
1	2	1	1	4	4	9	3	-0.3	0.7	0	1	0	1	1	-0.1
0	4	3	4	0	0	4	2	-0.2	0.8	1	9	4	14	3	-0.3
0	3	0	0	1	0	1	1	-0.1	0.9	1	4	1	6	2	-0.2
0	3	1	4	1	4	9	3	-0.3	0.7	1	4	0	5	2	-0.2
5	0	3	9	16	0	25	5	-0.5	0.6	16	1	4	21	4	-0.4
1	4	3	1	0	0	1	1	-0.1	0.9	0	9	4	13	3	-0.3

							b		
a_i	20	10	10	10	0	10			
$a_i\, y_i$							-1		
sign									

SVM

Linear

	x'_1	x'_2
x'_j	2	1
	4	1
	3	1

y_i		x_i					
1	x_1	1	2	1	13	2
-1	x_2	0	4	3	25	7
1	x_3	0	3	0	12	3
-1	x_4	0	3	1	15	4
1	x_5	5	0	3	19	8
-1	x_6	1	4	3	27	8

Kernel Matrix

$K(x_i, x'_j)$

							b		
a_i	2	0	1	1	1	1			
$a_i\, y_i$	2	0	1	-1	1	-1	2	17	5
sign								+	+

Decision Boundary

$b + \sum a_i\, y_i\, K$

RBF

x'_j	2	4	3															
x_i	1	2	1	$(x - x')^2$			$\sum$	$\sqrt{\ }$	$-\gamma\|\|^2$ (-0.1)	$e^{\square}$								
	1	4	4	1	4	4	9	3	-0.3	0.7	0	1	0	1	1	-0.1	0.9
	0	4	3	4	0	0	4	2	-0.2	0.8	1	9	4	14	3	-0.3	0.7
	0	3	0	0	1	0	1	1	-0.1	0.9	1	4	1	6	2	-0.2	0.8
	0	3	1	4	1	4	9	3	-0.3	0.7	1	4	0	5	2	-0.2	0.8
	5	0	3	9	16	0	25	5	-0.5	0.6	16	1	4	21	4	-0.4	0.67
	1	4	3	1	0	0	1	1	-0.1	0.9	0	9	4	13	3	-0.3	0.9

L2 distance $||^2$

$-\gamma||^2$	$e^{\square}$
-0.5	0.6
-0.4	0.67
-0.3	0.7
-0.2	0.8
-0.1	0.9

a_i	20	10	10	10	0	10		
$a_i\, y_i$							-1	
sign								

SVM

Linear

	x'_1	x'_2
x'_j	2	1
	4	1
	3	1

y_i		x_i					
1	x_1	1	2	1	13	2
-1	x_2	0	4	3	25	7
1	x_3	0	3	0	12	3
-1	x_4	0	3	1	15	4
1	x_5	5	0	3	19	8
-1	x_6	1	4	3	27	8

Kernel Matrix

$K(x_i, x'_j)$

							b		
a_i	2	0	1	1	1	1			
$a_i y_i$	2	0	1	-1	1	-1	2	17	5
sign								+	+

Decision Boundary

$b + \sum a_i y_i K$

RBF

	x'_1	x'_2
x'_j	2	1
	4	1
	3	1

L2 distance $||^2$

$-\gamma||^2$	$e^{\square}$
-0.5	0.6
-0.4	0.67
-0.3	0.7
-0.2	0.8
-0.1	0.9

x_i			$(x - x')^2$			$\sum$	$\sqrt{\ }$	$-\gamma||^2$ (-0.1)	$e^{\square}$				$\sqrt{\ }$	$-\gamma||^2$	$e^{\square}$
1	2	1	1	4	4	9	3	-0.3	0.7	0	1	0	1	-0.1	0.9
0	4	3	4	0	0	4	2	-0.2	0.8	1	9	4	14	-0.3	0.7
0	3	0	0	1	0	1	1	-0.1	0.9	1	4	1	6	-0.2	0.8
0	3	1	4	1	4	9	3	-0.3	0.7	1	4	0	5	-0.2	0.8
5	0	3	9	16	0	25	5	-0.5	0.6	16	1	4	21	-0.4	0.67
1	4	3	1	0	0	1	1	-0.1	0.9	0	9	4	13	-0.3	0.9

							b		
a_i	20	10	10	10	0	10			
$a_i y_i$	20	-10	10	-10	0	-10	-1		
sign									

SVM

Linear

	x'_1	x'_2
x'_j	2	1
	4	1
	3	1

y_i		x_i					
1	x_1	1	2	1	13	2
-1	x_2	0	4	3	25	7
1	x_3	0	3	0	12	3
-1	x_4	0	3	1	15	4
1	x_5	5	0	3	19	8
-1	x_6	1	4	3	27	8

Kernel Matrix

$K(x_i, x'_j)$

							b		
a_i	2	0	1	1	1	1			
$a_i y_i$	2	0	1	-1	1	-1	2	17	5
sign								+	+

Decision Boundary

$b + \sum a_i y_i K$

RBF

	x'_1	x'_2
x'_j	2	1
	4	1
	3	1

L2 distance $||^2$

$-\gamma||^2$	$e^{\square}$
-0.5	0.6
-0.4	0.67
-0.3	0.7
-0.2	0.8
-0.1	0.9

x_i			$(x - x')^2$			$\sum$	$\sqrt{\ }$	$-\gamma||^2$ (-0.1)	$e^{\square}$				$\sum$	$\sqrt{\ }$	$-\gamma||^2$	$e^{\square}$
1	2	1	1	4	4	9	3	-0.3	0.7	0	1	0	1	1	-0.1	0.9
0	4	3	4	0	0	4	2	-0.2	0.8	1	9	4	14	3	-0.3	0.7
0	3	0	0	1	0	1	1	-0.1	0.9	1	4	1	6	2	-0.2	0.8
0	3	1	4	1	4	9	3	-0.3	0.7	1	4	0	5	2	-0.2	0.8
5	0	3	9	16	0	25	5	-0.5	0.6	16	1	4	21	4	-0.4	0.67
1	4	3	1	0	0	1	1	-0.1	0.9	0	9	4	13	3	-0.3	0.9

							b		
a_i	20	10	10	10	0	10			
$a_i y_i$	20	-10	10	-10	0	-10	-1	-2	1
sign									

SVM

Linear

	x'_1	x'_2
x'_j	2	1
	4	1
	3	1

y_i		x_i					
1	x_1	1	2	1	13	2
-1	x_2	0	4	3	25	7
1	x_3	0	3	0	12	3
-1	x_4	0	3	1	15	4
1	x_5	5	0	3	19	8
-1	x_6	1	4	3	27	8

Kernel Matrix

$K(x_i, x'_j)$

							b		
a_i	2	0	1	1	1	1			
$a_i\, y_i$	2	0	1	-1	1	-1	2	→ 17	5
sign								+	+

Decision Boundary

$b + \sum a_i\, y_i\, K$

RBF

	x'_1	x'_2
x'_j	2	1
	4	1
	3	1

L2 distance $||^2$

$-\gamma||^2$	$e^{\square}$
-0.5	0.6
-0.4	0.67
-0.3	0.7
-0.2	0.8
-0.1	0.9

x_i			$(x - x')^2$			$\sum$	$\sqrt{\ }$	$-\gamma||^2$ (-0.1)	$e^{\square}$				$\sqrt{\ }$	$-\gamma||^2$	$e^{\square}$
1	2	1	1	4	4	9	3	-0.3	0.7	0	1	0	1	-0.1	0.9
0	4	3	4	0	0	4	2	-0.2	0.8	1	9	4	14	-0.3	0.7
0	3	0	0	1	0	1	1	-0.1	0.9	1	4	1	6	-0.2	0.8
0	3	1	4	1	4	9	3	-0.3	0.7	1	4	0	5	-0.2	0.8
5	0	3	9	16	0	25	5	-0.5	0.6	16	1	4	21	-0.4	0.67
1	4	3	1	0	0	1	1	-0.1	0.9	0	9	4	13	-0.3	0.9

							b		
a_i	20	10	10	10	0	10			
$a_i\, y_i$	20	-10	10	-10	0	-10	-1	→ -2	1
sign								−	+